23 Prove that $4x^2 - 8x + 7$ is positive for all values of x.

$4x^2 - 8x + 7 = 4(x^2 - 2x) + 7$

1 mark
Factor4 from quadratic terms.

$= 4[(x - 1)^2 - 1] + 7$

1 mark
Complete the square.

$= 4(x - 1)^2 + 3$

For every real x, $(x - 1)^2 \geq 0$.

So $4(x - 1)^2 + 3 \geq 3 > 0$.

1 mark
A square is non-negative; minimum is 3.

Therefore $4x^2 - 8x + 7$ is always positive.

(Total for Question 23 is 3 marks)